Ben (Pin-Chu) Cheng

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WORK EXPERIENCE

Changing The Present - Nonprofit Giving Platform **Remote**
Volunteer Analyst *Sep 2026 – Present*

- Automated intern welcome banners with an Abyssale variable-URL template, ending manual per-banner design.
- Built a Google Sheet as the intern data source, with formulas generating a shareable banner link for each new row.
- Reported progress in daily reports and coordinated requirements, data, and access directly with the co-founder.

UW Sensors, Energy & Automation Lab **Seattle, WA**
Graduate Research Assistant, Data Analysis *Jan 2026 – June 2026*

- Established a positive relationship between in-game engagement and GPA gains by fitting linear regression across ~10,000 interaction and assignment-completion records in SQL and Python (statsmodels).
- Exposed a 34% task-level drop-off concentrated in three assignment stages by tracking completion rate and session duration in Tableau, redirecting client milestone priorities.
- Standardized measurement across four engagement and UX research questions by authoring documented methods and metric definitions adopted into the team's 200-page client milestone deliverable.

Amazon x Extern **Seattle, WA**
Data Analyst Extern *Dec 2025 – Mar 2026*

- Unified 500+ employee reviews from Glassdoor, Indeed, Reddit, and YouTube into one Python (pandas) dataset, standardizing five fields and removing 12% duplicates before analysis.
- Isolated the two issue categories driving the most negative sentiment, running sentiment analysis, topic extraction, and rating-trend comparison across employee groups in Python (pandas, NLTK).
- Developed a 5-view interactive Power BI dashboard covering sentiment score, department satisfaction, rating trends, and theme distribution, filterable by department, time range, topic, and platform.

CollegeWay -Test-prep Education Firm **Taipei, Taiwan**
Data Analyst & Instructor *Sep 2020 – Aug 2024*

- Designed the firm's first data infrastructure, consolidating four sources (~5,000 mock-test scores, 100,000+ message logs, enrollment, profiles) into 5 SQL tables with reconciled student identities.
- Cut twice-weekly reporting from two days of manual compilation to ~1 hour by automating ETL from Excel (VBA, pivot tables) into SQL, owning validation.
- Launched the first Power BI dashboards on question-level results versus taught material, using SQL window functions to surface per-student trends and isolate instructional gaps.
- Drove lecture content restructuring affecting 200+ students annually by presenting monthly question-level findings to six stakeholders, including senior instructors at an organization lacking prior analytics.

RESEARCH PROJECTS

Medicare Hospital Cost Variation Analysis - IMT 600 Independent Study, University of Washington *Autumn 2026*

- Building a Databricks SQL pipeline joining ~146K CMS payment records to 5,400 hospitals, fixing a broken join key.
- Fitting a statsmodels regression to flag hospitals costing more than quality, ownership, and location explain.

Olist E-Commerce Delivery & Reviews Analysis - University of Washington

- Analyzed ~100,000 Brazilian e-commerce orders in Python (Pandas, NumPy) to root-cause one-star reviews: 8.1% of orders arrived late, and late orders drew 46% one-star reviews versus 6.6% for on-time, a 7x gap.
- Pinpointed worst states (Alagoas 24%, Maranhao 20% late); sized a 250-360 one-star review recovery in Rio de Janeiro.

Retail Sales Data Warehouse & Tableau Analytics - University of Washington

- Built a Snowflake star-schema sales warehouse with a SQL ELT pipeline: staging, deduplication, surrogate keys.
- Mapped $580M in sales in a 7-view Tableau Story; single-store states averaged $119.8M vs. $84.6M for multi-store.

SKILLS

SQL (window functions, joins, CTEs) | Python (pandas, NumPy, statsmodels, NLTK) | Excel (VBA) | Git | Databricks | Snowflake | Power BI | Tableau | linear regression | hypothesis testing | root-cause and funnel analysis | NLP sentiment analysis | ETL/ELT pipelines | star-schema modeling | data cleaning

EDUCATION

University of Washington, Master of Science in Information Management (MSIM), expected June 2027 GPA: 3.8
University of Washington, Bachelor of Arts in Communication, 2024 GPA: 3.8